

Homework 5

For Thursday, October 1, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. Limits are taken within the stated domain. You may use results covered through Lecture 6, including the derivative definition and the calculation of the derivative of x^2 . Do not use differentiation rules or L'Hospital's rule.

1. Let $f : \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R}$. Write the definition of

$$\lim_{x \rightarrow 2} f(x) = -\infty$$

using M and δ , and then write its logical negation.

2. Suppose f is defined for $0 < |x| < 1$ and satisfies

$$|f(x) - (1 + 2x)| \leq 3x^2.$$

Determine $\lim_{x \rightarrow 0} f(x)$. Prove your answer directly from the epsilon-delta definition, giving an explicit δ for every $\varepsilon > 0$.

3. Does the condition $0 \leq g(x) \leq 1$ for $0 < |x| < 1$ force $\lim_{x \rightarrow 0} g(x)$ to exist? Give one function satisfying the condition whose limit exists and another satisfying it whose limit does not exist. Justify both conclusions.

4. Evaluate

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{\tan(2x)}.$$

5. Find $c \in \mathbb{R}$ such that the function

$$F(x) = \begin{cases} \frac{1 - \cos x}{x^2}, & x \neq 0, \\ c, & x = 0 \end{cases}$$

is continuous at zero.

6. Prove directly from the definition that

$$\lim_{x \rightarrow 1} \frac{2 + \sin x}{(x - 1)^2} = +\infty.$$

Give an explicit δ for every $M > 0$.

7. For $x \neq 0$, let

$$f(x) = \frac{1 + \sin(1/x)}{x^2}.$$

Prove that f is unbounded above on every deleted interval $0 < |x| < \delta$, but does not tend to $+\infty$ as $x \rightarrow 0$. *Hint:* consider points for which $1/x = \pi/2 + 2\pi n$ or $1/x = 3\pi/2 + 2\pi n$, where n is a positive integer.

8. Consider the function

$$r(x) = \frac{x^2 - 1}{(x - 1)(x + 2)}$$

on its natural real domain. At each excluded point, determine whether the graph has a hole or a vertical asymptote. Give the finite limit at a hole and the two one-sided limits at a vertical asymptote. Also find every horizontal asymptote.

9. Find both limits

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{9x^2 + 1}}{x}, \quad \lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2 + 1}}{x}.$$

Explain why the two answers differ.

10. Prove directly from the definition that

$$\lim_{x \rightarrow +\infty} \frac{2x + 1}{x - 3} = 2.$$

For every $\varepsilon > 0$, give an explicit $A > 0$ such that the required inequality holds for every $x > A$.

11. Find an oblique asymptote $y = mx + b$ of

$$f(x) = \frac{x^2 + 3x + 4}{x + 1}, \quad x \neq -1.$$

Verify it at both ends by computing the limits of $f(x) - (mx + b)$ as $x \rightarrow +\infty$ and $x \rightarrow -\infty$.

12. Construct a rational function with domain $\mathbb{R} \setminus \{1\}$, vertical asymptote $x = 1$, horizontal asymptote $y = 2$ at both ends, and value $f(0) = 0$. Give a formula and verify all four properties.
13. Let $f(x) = \sqrt{x}$, with domain $[0, \infty)$. Use the definition of the derivative to calculate $f'(4)$. Then give the equation of the tangent line at $(4, 2)$.

14. Suppose f is defined on an open interval containing 3, with $f(3) = 7$ and $f'(3) = -2$. Find

$$\lim_{h \rightarrow 0} \frac{f(3 + 4h) - 7}{h}.$$

Explain how the derivative definition gives your answer.

15. Find all tangent lines to the parabola $y = x^2$ that pass through the point $(0, -4)$. Give each line's equation and point of tangency, and explain why there are no others.